

ton. Their studies confirmed that in money matters, too, most people exaggerate the likelihood of occurrences. If something can happen, then it must be a concern, whether the probability of the event is one in a thousand or one in a million. This explains the popularity of cancer insurance, flight insurance, and other forms of protection against catastrophes, even though the policies cost far more than an actuarially fair price.

People overestimate the value of a chance on a long-shot win just as much as they blow up the dangers of rare, spectacular events. Gambling casinos, racetracks, and lotteries are always busy. There is a steady flow of wealth from hopeful gamblers to the men who know what the odds really are.

This phenomenon is verifiably true in the case of call options on stocks. Very-low-priced options sell for higher prices than their valuations based on probability theory would suggest, because they are like lottery tickets. Sensible investors should avoid, too, very-low-priced, very risky stocks. Two-dollar "bargain" stocks are probably overpriced, despite—or because of—their very low price. W. C. Fields, in *The Bank Dick*, hit it rich in Beefsteak Mines stock, but long-shot investments are poor bets in real life.

One of Tversky and Kahneman's major discoveries was that a dollar of gain is not equal to a dollar lost in most people's minds, certainly not when the dollar stake is high. Who would bet his car against a neighbor's similar car at even odds? Very few, because having no car is more of a bad than having two cars is a good. Formally, the value function is nonlinear, so that individuals have more displeasure in losing a sum than the pleasure associated with winning the same amount. Most people want a two-thirds chance of winning to venture a big bet.

If most investors act in this risk-averse way, we can understand why they will accept a lower return from bonds and other fixed-income investments rather than hold stocks, and why they will stick with the blue chips instead of taking a chance on capturing the